

Data Preparation

Ting Kai Ming
Nanjing University

Reading

- Required reading: Aggarwal Chapter 2
- Supplementary reading: Tan Chapter 2 (excluding 2.4)

Acknowledgement:

Unless stated otherwise, the contents and images on the following slides are sourced from C. C. Aggarwal, Data Mining: The Textbook. Springer, 2016.

For the following eight discussion questions, refer to the materials presented by Tan Chapter 2 or in the 2-2_data powerpoint slides.

Discussion

1. *What is the attribute type of each of the following kinds of attributes (a) Age, (b) Salary, (c) Zip Code, (d) Province of Residence, (e) Height, (f) Weight.*
2. *An analyst sets up a sensor network in order to measure the temperature of different locations over a period. What is the data type of the data collected?*
3. *An analyst processes Web logs in order to create records with the ordering information for Web page accesses from different users. What is the type of this data?*
4. Describe the difference between interval and ratio attribute type.

Discussion

5. Suppose that you are employed as a data mining consultant for an Internet search engine company. Describe how data mining can help the company by giving specific examples of how techniques, such as clustering, classification, association rule mining, and anomaly detection can be applied.

Discussion

6. Distinguish between noise and outliers. Consider the following questions.

(a) Is noise ever interesting or desirable? Outliers?

(b) Can noise objects be outliers?

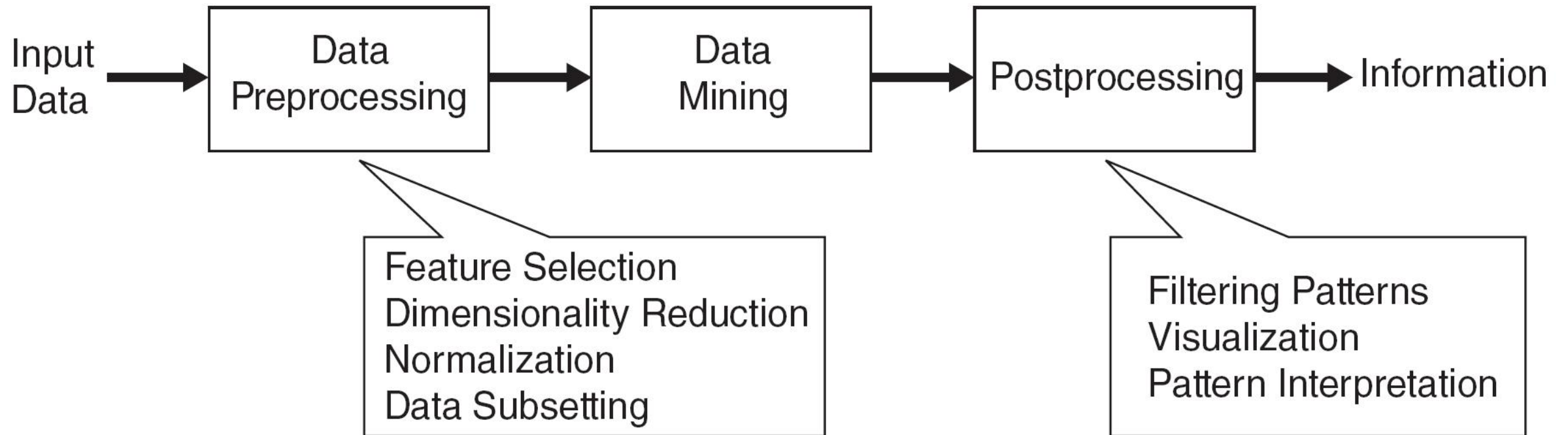
(c) Are noise objects always outliers?

(d) Are outliers always noise objects?

(e) Can noise make a typical value into an unusual one, or vice versa?

Discussion

7. The following attributes are measured for members of a herd of Asian elephants: *weight*, *height*, *tusk length*, *trunk length*, and *ear area*. Based on these measurements, what sort of similarity measure would you use to compare or group these elephants? Justify your answer and explain any special circumstances.
8. Discuss why a document-term matrix is an example of a data set that has asymmetric discrete or asymmetric continuous features.



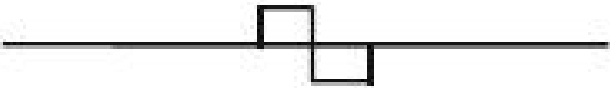
Converting from one data type to another

Source data type	Destination data type	Methods
Numeric	Categorical	Discretization
Categorical	Numeric	Binarization
Text	Numeric	Latent semantic analysis (<i>LSA</i>)
Time series	Discrete sequence	<i>SAX</i>
★ Time series	Numeric multidimensional	<i>DWT, DFT</i>
Discrete sequence	Numeric multidimensional	<i>DWT, DFT</i>
★ Spatial	Numeric multidimensional	2-d <i>DWT</i>
Graphs	Numeric multidimensional	<i>MDS</i> , spectral
Any type	Graphs	Similarity graph (Restricted applicability)

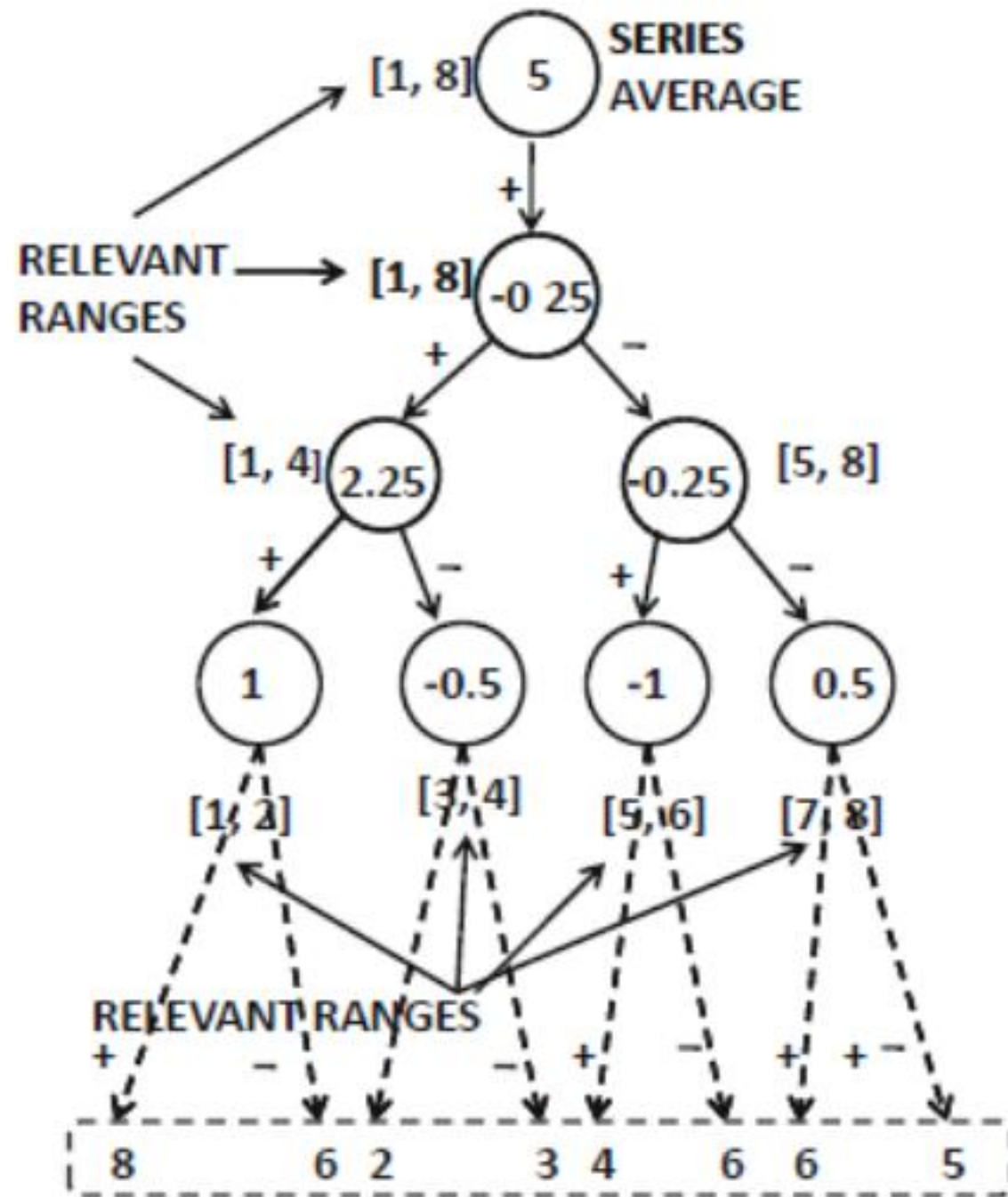
Haar Wavelet Transform

Wavelet decomposition of a time series S is provided in two steps:

1. Report half the average difference of the behavioral attribute values between the first and second temporal halves of S as a wavelet coefficient.
 2. Recursively apply this approach to first and second temporal halves of S .
- *The wavelet representation is a decomposition of the original time series of length q into the weighted sum of a set of q “simpler” time series (or wavelets) that are orthogonal to one another.*
 - *These “simpler” time series are the basis vectors, and the wavelet coefficients represent the weights of the different basis vectors in the decomposition.*

SERIES AVERAGES	WAVELET COEFFICIENT	WAVELET SHAPE	BASIS VECTOR
(8, 6, 2, 3, 4, 6, 6, 5)	1		1 -1 0 0 0 0 0 0
	-0.5		0 0 1 -1 0 0 0 0
	-1		0 0 0 1 -1 0 0 0
	0.5		0 0 0 0 0 1 -1 0
(7, 2.5, 5, 5.5)	2.25		1 1 -1 -1 0 0 0 0
	-0.25		0 0 0 1 1 -1 -1 0
(4.75, 5.25)	-0.25		1 1 1 1 -1 -1 -1 -1
(5)	5		1 1 1 1 1 1 1 1

Error tree

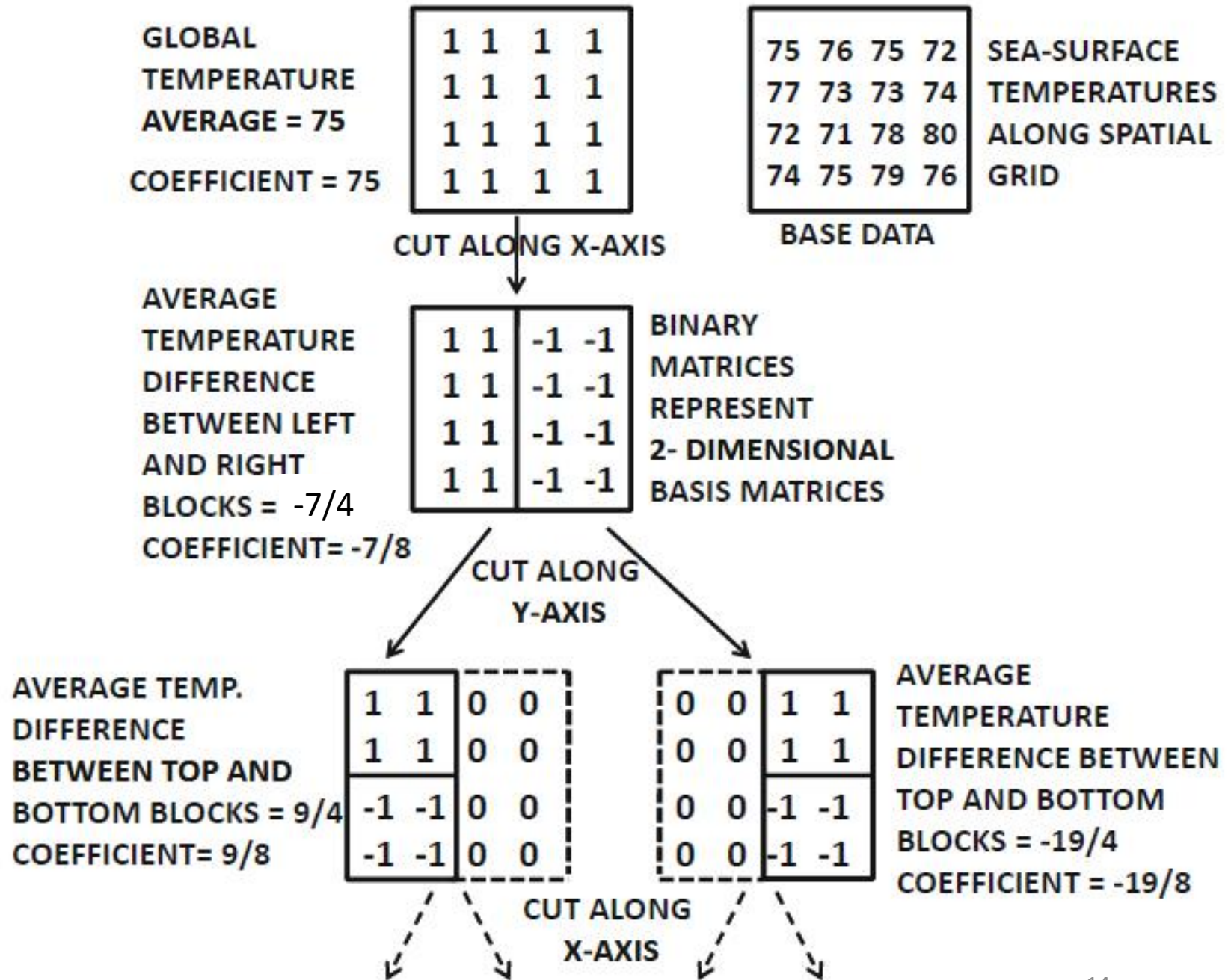


Convert a database of N time series into N multidimensional vectors.

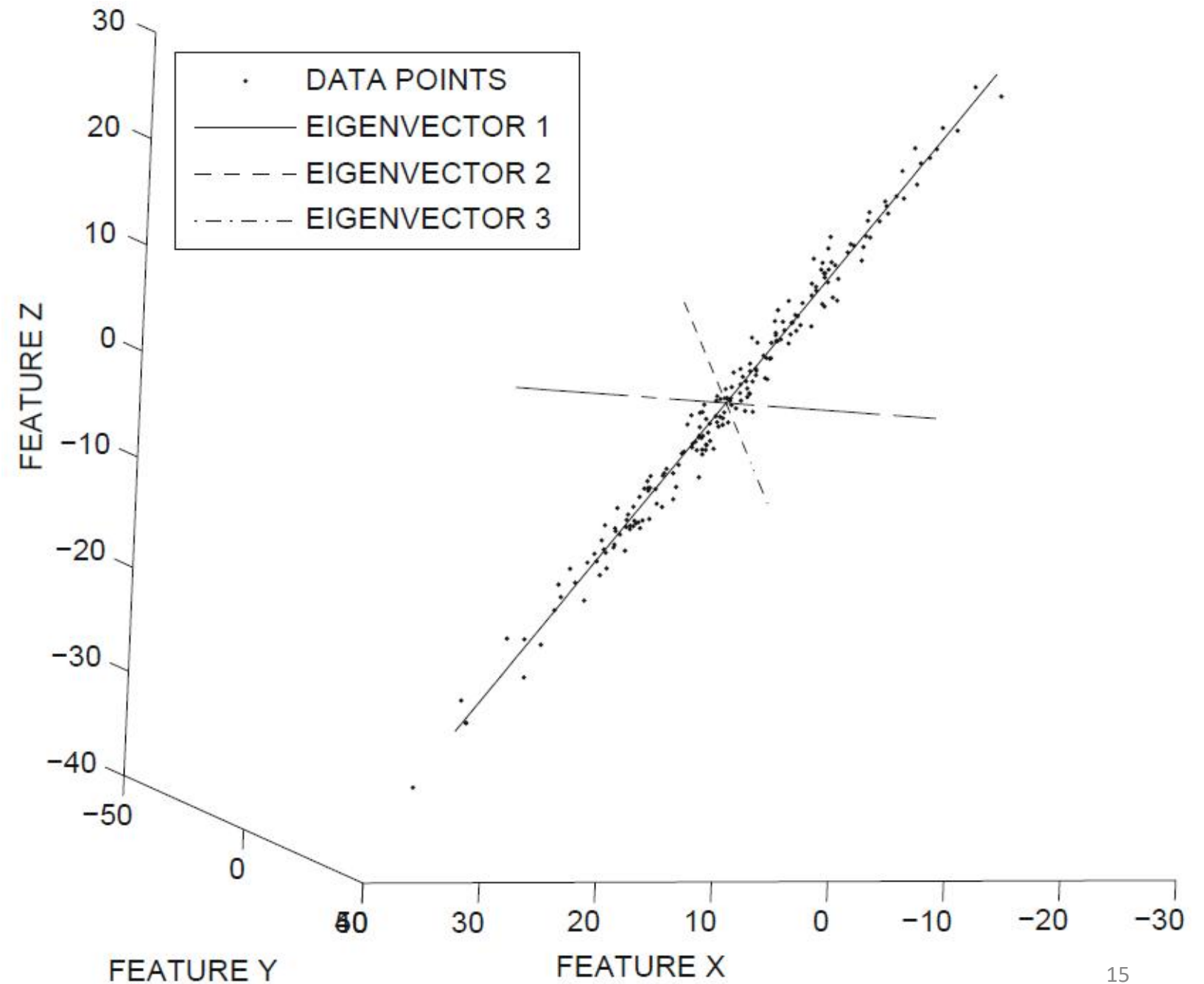
Two possible ways:

1. The coefficient for the same basis vector is selected for each series to create a meaningful multidimensional database of low dimensionality. Therefore, the basis vectors that have the largest *average* normalized coefficient across the N different series are selected.
2. The full dimensionality of the wavelet coefficient representation is retained. However, for each time series, the largest normalized coefficients (in magnitude) are selected individually. The remaining values are set to 0. This results in a sparse database of high dimensionality, in which many values are 0. A method such as *SVD* can be applied as a second step to further reduce the dimensionality.

Spatial to Multidimensional Transformation with Wavelets (Example)



Dimension Reduction: Example



Dimension reduction methods

- **Principal Component Analysis (*PCA*)**
- **Singular Value Decomposition (*SVD*)**
- **Latent Semantic Analysis**
 - an application of the *SVD* method to the text domain
- Other applications of *PCA* & *SVD*: Noise reduction, data imputation, matrix inversion, matrix algebra such as powers of a matrix.

SVD and *PCA* are extraordinarily useful because matrix and linear algebra operations are ubiquitous in data mining. *SVD* and *PCA* facilitate such matrix operations by providing convenient decompositions and basis representations.

Dimensionality reduction with type transformation

1. *Time series to multidimensional:* A number of methods, such as the discrete Fourier transform and discrete wavelet transform are used. While these methods can also be viewed as a rotation of an axis system defined by the various time stamps of the contextual attribute, the data are no longer dependency oriented after the rotation.
2. *Weighted graphs to multidimensional:* Multidimensional scaling and spectral methods are used to embed weighted graphs in multidimensional spaces, so that the similarity or distance values on the edges are captured by a multidimensional embedding.

Discussion

1. *Suppose that you had a set of arbitrary objects of different types representing different characteristics of widgets. A domain expert gave you the similarity value between every pair of objects. How would you convert these objects into a multidimensional data set?*

Discussion

2. *Consider the time-series 1, 1, 3, 3, 3, 3, 1, 1. Perform wavelet decomposition of the time series. How many coefficients of the series are non-zero?*
3. *Suppose that you had a data set, such that each data point corresponds to sea surface temperatures over a square kilometer of resolution 10×10 . In other words, each data record contains a 10×10 grid of temperature values with spatial locations. You also have some text associated with each 10×10 grid.
How would you convert this data into a multidimensional data set?*

Discussion

4. *Suppose that you had a set of discrete biological protein sequences, which are annotated with text describing the properties of the protein. How would you create a multidimensional representation from this heterogeneous data set?*
5. *Apart from the assumption that the specified distance matrix is Euclidean, what is the assumption of using Multidimensional Scaling (MDS) in converting a graph to multidimensional representation? Hint: The violation of the assumption will not allow MDS to convert a graph.*

Converting from one data type to another

Source data type	Destination data type	Methods
Numeric	Categorical	Discretization
Categorical	Numeric	Binarization
Text	Numeric	Latent semantic analysis (<i>LSA</i>)
Time series	Discrete sequence	<i>SAX</i>
Time series	Numeric multidimensional	<i>DWT, DFT</i>
Discrete sequence	Numeric multidimensional	<i>DWT, DFT</i>
Spatial	Numeric multidimensional	2-d <i>DWT</i>
★ Graphs	Numeric multidimensional	<i>MDS</i> , spectral
★ Any type	Graphs	Similarity graph (Restricted applicability)

Any Type to Graphs for Similarity-Based Applications

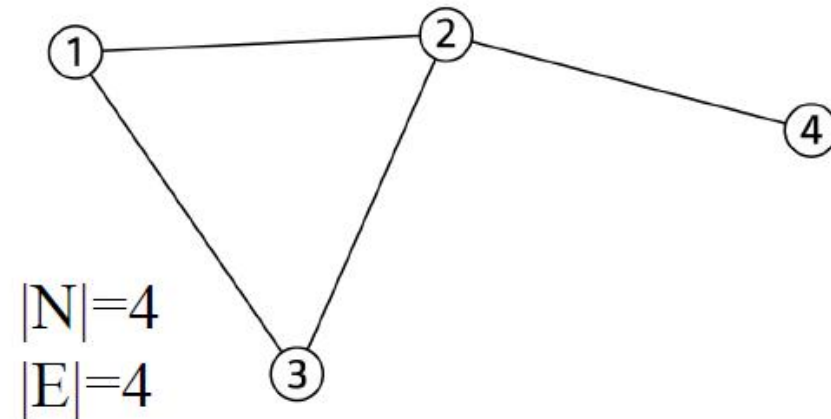
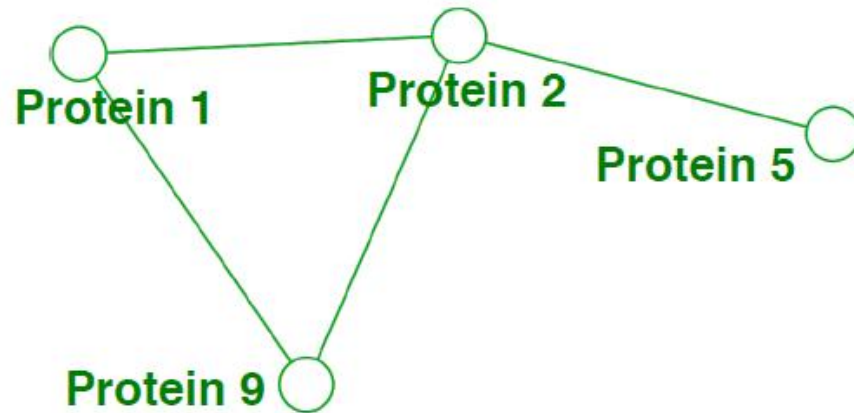
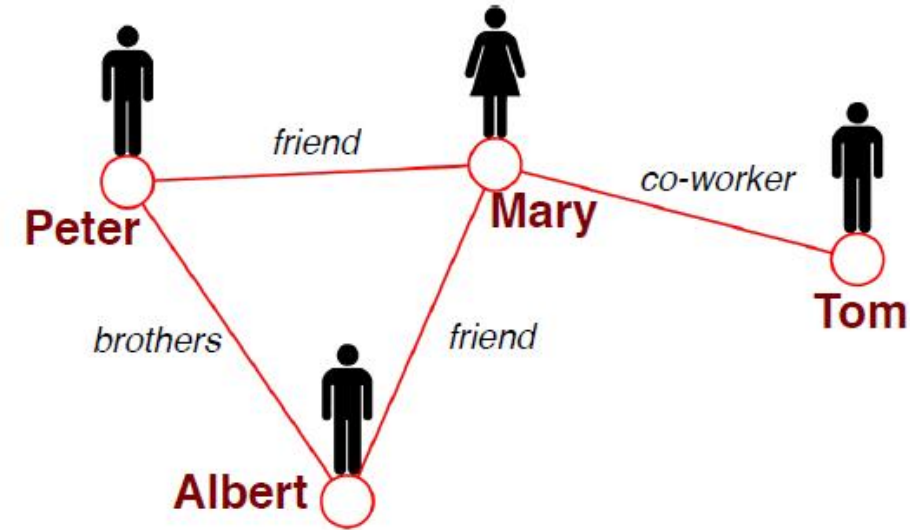
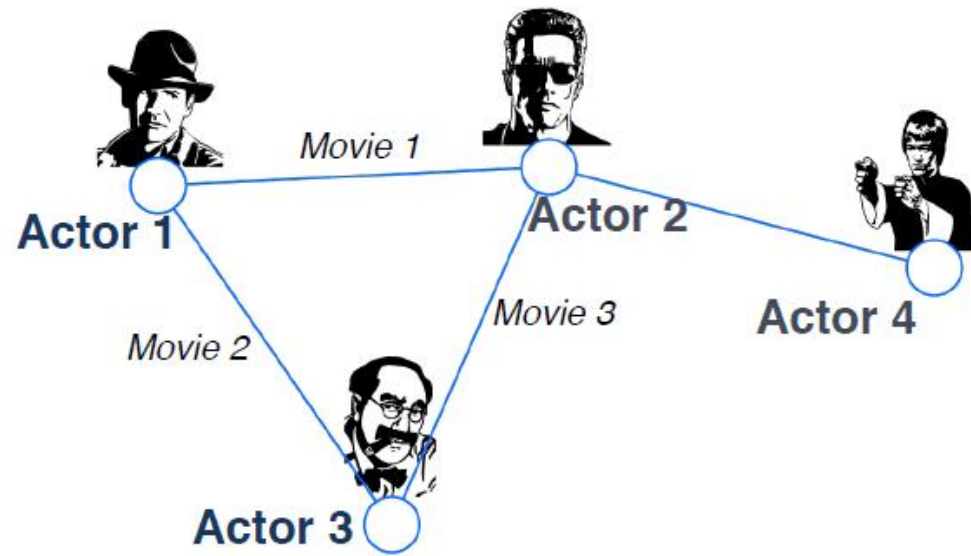
For a given set of data objects $D = \{O_1 \dots O_n\}$, a neighborhood graph is defined as follows:

1. A single node is defined for each object in D . This is defined by the node set N , containing n nodes where the node i corresponds to the object O_i .
2. An edge exists between O_i and O_j , if the distance $d(O_i, O_j)$ is less than a particular threshold. Alternatively, the k -nearest neighbors of each node may be used. Because the k -nearest neighbor relationship is not symmetric, this results in a directed graph. The directions on the edges are ignored, and the parallel edges are removed. The weight w_{ij} of the edge (i, j) is equal to a kernelized function of the distance between the objects O_i and O_j , so that larger weights indicate greater similarity. An example is the *heat kernel*:

$$w_{ij} = \exp[-d(O_i, O_j)^2/t^2]$$

where t is a user-defined parameter.

Example complex data objects: Graph/network



Spectral Transformation and Embedding of Graphs

- Spectral methods work with *similarity graphs* in which the weights on the edges represent similarity rather than distances.
- It transforms a similarity graph to a multidimensional representation that preserves the *local similarity structure* between nodes.
- Unlike *MDS*, which works with a complete graph of *global* distances, this graph is generally a *sparsified* representation of the similarity of each object to its *K nearest objects*.
- It is desired to embed the nodes of this graph into a *k*-dimensional space so that the similarity structure of the data is preserved.
- The problem is formulated as an eigen decomposition of similarity matrix that results in *k* eigenvectors with eigenvalues $\lambda_2 \leq \lambda_3 \leq \dots \leq \lambda_{k+1}$.
- This corresponds to the *k*-dimensional embedding of the *n* nodes.

Applications of spectral transformation

Any data type → similarity graph → graph embedding in
multidimensional representation

- Typical application: spectral clustering.
 - Spectral clustering usually employs k-means clustering at the final stage to perform the actual clustering.
- Though it may be applied for other data mining tasks, its efficacy is yet to be demonstrated.

Discussion

1. What is the assumption of spectral methods?
2. Describe how you would perform clustering using spectral clustering if the given dataset is in
 - a. Multidimensional representation
 - b. Time series
 - c. Graphs
3. *MDS* methods are designed for preserving global distances; and spectral methods are designed for preserving local distances. Explain the meanings of global distances and local distances in this context.

Week 2 Summary

- The topic is data preparation. There are many ways to prepare a given dataset of any data type before it is applied to a data mining algorithm.
- We have focused on methods for data type conversion and dimension reduction. Other data preparation includes missing value imputation, normalization and sampling (in order to reduce data size).
- A key approach is to convert a given dataset of complex data type into multidimensional data (with the iid assumption) so that many existing data mining methods could be applied for this dataset.

Reflection

- What have you learned this week?
(Every student shall attempt to answer this question.)

